Technological Forecasting & Social Change 80 (2013) 789–800

Contents lists available at SciVerse ScienceDirect

Technological Forecasting & Social Change

Dynamic scenario discovery under deep uncertainty: The future of copper

Jan H. Kwakkel [a,*], Willem L. Auping [b], Erik Pruyt [a]

[a] Faculty of Technology, Policy & Management, Delft University of Technology, Delft, The Netherlands
[b] The Hague Centre for Strategic Studies, The Hague, The Netherlands

ARTICLE INFO

Article history:
Received 3 December 2011
Received in revised form 11 September 2012
Accepted 22 September 2012
Available online 30 October 2012

Keywords:
Scenario discovery
Exploratory modeling and analysis
System dynamics
Deep uncertainty
Metal scarcity

ABSTRACT

Scenarios are commonly used to communicate and characterize uncertainty in many policy fields. One of the main challenges of scenario approaches is that analysts have to try and capture the full breadth of uncertainty about the future in a small set of scenarios. In the presence of deep uncertainty, this is even more challenging. Scenario discovery is a model-based technique inspired by the scenario logic school that addresses this challenge. In scenario discovery, an ensemble of model runs is created that encompasses the various uncertainties perceived by the actors involved in particular decision making situations. The ensemble is subsequently screened to identify runs of interest, and their conditions for occurring are identified through machine learning. Here, we extend scenario discovery to cope with dynamics over time. To this end, a time series clustering approach is applied to the ensemble of model runs in order to identify different types of dynamics. The types of dynamics are subsequently analyzed to identify dynamics that are of interest, and their causes for occurrence are revealed. This dynamic scenario discovery approach is illustrated with a case about copper scarcity.

© 2012 Elsevier Inc. All rights reserved.

1. Introduction

Scenarios provide a commonly used means to communicate and characterize uncertainty in many decision support applications. There exists a plethora of scenario definitions, typologies, and methodologies [1,2]. A distinction can be made between the *La Prospective* school developed in France, the Probabilistic Modified Trends school originated at RAND, and the Intuitive Logic school typically associated with the work of Shell [2]. In the evaluative literature, one of the reported problems of traditional scenario approaches is that they often struggle in case of problems that involve a variety of actors with quite diverse world views [3] or when there is a lacking consensus [4]. Scenario approaches also struggle with anticipating rare events [5] and grapple with the multiplicity of plausible futures [6]. The challenge of traditional scenario approaches is that analysts have to try and capture the full breadth of the uncertainty about the future in a small set of scenarios that need to be intelligible and useful to both the actors involved in the scenario development process and analysts supporting this process [7–9]. Developing or identifying a handful of scenarios, that fully represent all plausible futures is difficult. Communicating, and using more than a handful of representative scenarios is equally difficult and may even be counterproductive [9]. The intuitive logic school addresses these problems through the identification of the factors that are both highly uncertain and can have a profound impact on the decision problem at hand [8]. However, this works mainly if the group of involved actors is relatively small, their interests and concerns are known, and overlap to a certain extent [2]. Moreover, how to best represent the diversity contained in all the uncertain factors in a small set of scenarios, is a continuing challenge [10].

Recently, an approach called scenario discovery [7,10,11] has been put forward as a technique that can be used for developing scenarios for problems that involve a large number of actors with quite diverging world views and values and where there are many uncertain factors. Scenario discovery is a model driven approach that builds on the intuitive logic school [7]. Scenario

* Corresponding author.
E-mail addresses: J.H.Kwakkel@tudelft.nl (J.H. Kwakkel), willemAuping@hcss.nl (W.L. Auping), E.Pruyt@tudelft.nl (E. Pruyt).

0040-1625/$ – see front matter © 2012 Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.techfore.2012.09.012

discovery builds on earlier work on using models for decision making under deep uncertainty [12–14]. It starts from an ensemble of model runs that is analyzed in order to identify runs that are of particular interest. Next, these runs of interest are analyzed to reveal the combinations of factors responsible for generating them. The documented cases of scenario discovery have used a single model with a small set of uncertain parameters as the basis for generating the ensemble of runs. For example [7], uses a model with 8 uncertain parameters, and [10] uses a model with 20 uncertain parameters, and the identification of interesting runs in both cases is based on the terminal values of each individual run of outcome indicators related to policy performance [7,10].

In this paper, we extend the scenario discovery approach conceptually, technically, and practically. Conceptually, we understand scenarios not as states of the world but as developments over time. Technically, this implies that the machine learning techniques usually applied in scenario discovery cannot be applied straightforwardly. To overcome this problem, we use time series clustering for the identification of sets of behaviors over time, thus transforming time series results to scalar values that can be used as input to the various machine learning techniques that can be used for scenario discovery. Practically, we extend scenario discovery by working with two structurally distinct models that share only a subset of the uncertain factors, and jointly cover significantly more uncertain parameters than earlier applications of scenario discovery. These practical extensions pose additional challenges in the design of the computational experiments and the analysis of the results.

To illustrate our extended scenario discovery approach, we apply it to the problem of copper scarcity. There has been a growing attention to mineral and metal scarcity, but this attention has been focused mainly on lithium, rare earth metals and other metals characterized by supply risks due to the limited number of countries where it is mined. However, bulk metals can also suffer from scarcity, as evidenced by the copper price which has been on a high level since 2005 [15], resulting in phenomena like the theft of copper wiring. Crisis behavior in the copper market may have profound impacts on society beyond increased copper theft, and may be particularly worrisome with regard to a transition towards more sustainable energy systems [16]. The main aim of the case was therefore to identify the various ways in which the copper system – composed of supply, demand, recycling, and substitution – could evolve, the kinds of dynamics that could occur, the undesirable price dynamics, and the causes for their occurrence.

The physical side of the copper system is well documented [e.g. 17,18] and does not contain much uncertainty. However, with respect to the way in which demand should be represented, there are profoundly diverging views: there are those who argue that copper demand should be modeled at a high level of aggregation as a function of world population, while others argue that one should use a bottom up approach from the various types of usages to the overall demand [19–21]. As argued by Cole, "whether a 'top-down' or 'bottom-up' approach is chosen, however, may affect the results. Simple recursive calculation of global or regional aggregates broken down by sector often gives surprisingly different results from systematically building up the global or regional aggregates from the sector or subsector levels" [22]. Other sources of uncertainty are the development of the ore grade [17,23], the impacts of substitution behavior [24], and various geopolitical developments, such as the growing copper demand in developing economies [17,25]. The various uncertainties are captured in two distinct simulation models. One represents a bottom up modeling of demand, while the other represents a top down modeling of demand. The supply system is essentially the same in both models. The behavior of these models is explored across a wide range of parametric uncertainties using Latin Hypercube Sampling. The results are clustered using a time series clustering approach, and subsequently analyzed using the patient rule induction method (PRIM) [26], a particular machine learning technique. Exemplars of undesirable dynamics are identified, and their conditions for occurring derived.

In the next section, we review the current scenario discovery approach and outline where and how we have extended it to cope with dynamics over time. In Section 3, we illustrate this modified scenario discovery approach through the case of copper scarcity. Section 4 discusses these results from a methodological point of view. Section 5 presents the main conclusions.

2. Dynamic scenario discovery

Scenario discovery addresses problems encountered when trying to develop model-based scenarios for problems that involve a large number of actors with diverging world views and values, or that are characterized by a very large number of uncertain factors. Typical for such problems is that the analysts do not know, or the parties to a decision cannot agree on (1) the appropriate conceptual models that describe the relationships among the key driving forces that shape the long-term future, (2) the probability distributions used to represent uncertainty about key variables and parameters in the mathematical representations of these conceptual models, and/or (3) how to value the desirability of alternative outcomes [14]. This is also called decision making under deep uncertainty, or severe uncertainty [14,27]. In the presence of a lack of knowledge or disagreement related to the model representation of a system and the evaluation of outcomes, enumeration of multiple alternatives for how (aspects of) the system works or are to be parameterized and how to value outcomes may still be possible, without being able to rank order these alternatives in terms of how likely or plausible they are judged to be [28].

2.1. Exploratory modeling and analysis

Scenario discovery builds on earlier work on using models for decision making under deep uncertainty [12–14]. Under deep uncertainty, it is not possible to develop a single model that accurately represents the system of interest. Exploratory modeling and analysis (EMA) [12] provides an alternative way of using the available information, data, and knowledge. An ensemble of models, consistent with the available knowledge, data, and information is developed. A single model run drawn from this

potentially infinite ensemble of models provides a computational experiment that reveals how the world would behave if the various guesses that particular model makes about the various uncertainties were correct. The behavioral landscape of this ensemble is explored using a series of computational experiments, and the behavioral landscape is analyzed using a variety of machine learning techniques. Scenario discovery can be understood as a particular application of EMA, where one tries to identify the combinations of uncertainties that produce regions of interest in the behavioral landscape of model outcomes. Typically, in scenario discovery, one looks for areas that represent vulnerabilities to proposed policies by looking at the terminal values of various outcomes of interest for the individual runs [7]. There is, however, no theoretical or conceptual reason why other criteria cannot be used for defining the regions of interest, nor is it necessary to restrict the identification of regions of interest to terminal values of outcomes of interest for the individual runs.

Important steps in EMA are to (i) conceptualize the decision problem and the associated uncertainties; (ii) develop an ensemble of fast and simple models of the system of interest; (iii) specify the uncertainties that are to be explored. Next, depending on the purpose for which EMA is applied, various subsequent steps are possible. In the case of scenario discovery, the typical subsequent steps are to (iv) analyze the behavioral landscape resulting from (iii); (v) identify the combinations of uncertainties from which regions of interest in the behavioral landscape originate; (vi) assess these combinations of uncertainties using various model quality metrics and related machine learning techniques for assessing model quality [7]; (vii) qualitatively or quantitatively communicate the typical futures in these regions of interest, i.e. exemplary scenarios, and the combinations of uncertainties from which the regions of interest in the behavioral landscape originate to the actors involved in the decision making problem. Based on these scenarios and regions, in the interaction with these actors, a process of policy formulation and reformulation can start. Candidate policies can be tested by iterating through the previous steps.

2.2. Algorithms for dynamic scenario discovery

Both the analysis of the behavioral landscape and the identification of uncertainties from which particular regions of interest in the behavioral landscape originate can utilize various machine learning techniques. In this paper, we are interested in dynamics over time. Thus, we analyze the behavioral landscape using time series clustering. From this analysis emerges a set of dynamics, some of which are judged to be of interest. In order to identify the combinations of uncertainties responsible for generating the behavior of each cluster of interest, we utilize a modified version of PRIM [26], which is typically used in scenario discovery [7,13].

2.2.1. Coping with dynamics: time series clustering

The goal of clustering is to organize an unlabeled data set into homogenous groups, in which the similarity within the group is minimized and the dissimilarity between groups is maximized [29,30]. Typically, clustering approaches are applied to static data [29]. Static clustering approaches can be divided into five families: partitioning methods, hierarchical methods, density based methods, grid-based methods, and model-based methods [31]. In general, time series clustering approaches try to modify existing clustering approaches for static data so that they can cope with time series data. Either the algorithm is modified to deal with the raw time series data, or the time series are processed in such a way that static clustering methods can be used directly. A substantial portion of the research on time series clustering focuses on modifying the similarity measure used in a clustering method to handle time series data [32]. To assess the efficacy of hierarchical clustering methods in capturing similarity, the use of dendrograms is recommended [32]. A review of the state of the art in time series clustering can be found in [29].

In this paper, we adopt an agglomerative hierarchical clustering approach. That is, we start by positioning each time series in its own cluster, and then hierarchically merge each cluster into larger and larger clusters [29]. The advantage of this approach is that it produces an ordering of similarity. The user can experiment with different similarity thresholds, producing a different number of final clusters, which aids the discovery of the types of dynamics.

2.2.2. Identifying undesirable regions: the patient rule induction method (PRIM)

After identifying regions of interest in the behavioral landscape of model outcomes, one wants to identify where in the model input space these regions of interest originate. That is, one wants to close the loop from uncertainties to behavior of interest and from behavior of interest back to the uncertainties. For this final step, PRIM is typically used in scenario discovery [7,10,13].

PRIM can be used for data analytic questions, where the analyst tries to find combinations of values for input variables that result in similar characteristic values for the outcome variables. Specifically, one seeks a set of subspaces of the model input space within which the values of the output variables are considerably different from their average values over the entire domain. PRIM describes these subspaces in the form of 'boxes' of the model input space. This results in a very concise representation, for typically only a limited set of dimensions of the model input space is restricted. That is, a subspace is characterized by upper and/or lower limits on only a few input dimensions. Still, interpretation of such a PRIM box can be challenging for the analyst because of the multi-dimensional character of the subspace. That is, in interpreting the results one has to account for all the restricted dimensions simultaneously. Another issue with PRIM is that the subspaces found by PRIM can overlap, further hampering interpretability. Although the original version of PRIM requires that the input data are continuous or categorical [26], it can be modified to deal with ordinal data [33].

The default objective function used by PRIM aims at maximizing the mean of the cases inside a box. The algorithm generates a set of candidate boxes and then selects that candidate box that maximizes this mean. In the case of categorical or ordinal data, there is a problem with this approach, for the various candidate boxes do not contain the same amount of data. Thus, the

comparison is biased. Moreover, this defeats one of the principal strengths of PRIM, namely its patient (or lenient) character, expressed in removing or adding only a few cases at a time. To address this problem, we modified the objective function used by PRIM. Instead of selecting the candidate box that maximizes the mean, we modified the algorithm to select the box where the increase in the mean divided by the change of the cardinality of the cases inside the box is largest. Thus, the gain in the mean is offset against the change in the number of cases inside a box.

In recent work on using PRIM for scenario discovery, it was shown that the results of PRIM could be improved by applying a preprocessing step based on principal components analysis (PCA) [34]. This preprocessing step involves identifying the experiments of interest. Next, the covariance matrix of these experiments is derived [35]. Then, the eigenvectors and eigenvalues of this covariance matrix are identified through singular value decomposition. The eigenvectors are sorted based on the eigenvalues. The resulting eigenspace is subsequently used for rotating all the experiments. That is, the experiments are rotated to the eigenspace of the covariance matrix of the experiments of interest. To ease interpretation of the results, instead of rotating the uncertain parameters jointly, one can cluster the uncertain parameters, and derive a rotation for each cluster separately. This extension of PRIM, known as PCA-PRIM, is applied here. We exclude categorical dimensions from this rotation.

Some computational support for scenario discovery exists. The Comprehensive R Archive Network (CRAN) contains a scenario discovery toolkit package. However, this implementation of PRIM does not handle ordinal data and categorical data correctly, and the pasting phase of the algorithm is not consistent with the algorithm as described by [26]. It is not uncommon to encounter categorical or ordinal data in scenario discovery. In scenario discovery, the model input space is defined by the uncertainties explored using EMA. Not all these uncertainties will be real valued. For example, if there is uncertainty about (an aspect of) model formulation, this can result in multiple distinct model formulations, which are distinct categories. We therefore re-implemented the PRIM algorithm to handle the three data types correctly. In this revised implementation we modified the pasting part of the algorithm to be consistent with [26]. This re-implementation also allowed us to modify the objective function used in PRIM.

3. Illustrating dynamic scenario discovery: the case of copper scarcity

The aim of this case study is to explore how the copper price can develop in the future and, through dynamic scenario discovery, identify plausible undesirable future dynamics of the copper price. To this end, we first conceptualize the copper system and the main uncertainties affecting the future evolution of the copper price. Second, we present an ensemble of system dynamics models that can be used to explore over these uncertainties. Third, we discuss the design of experiments to explore the behavior of this ensemble. Fourth, we apply the outlined dynamic scenario discovery approach to the results of the computational experiments.

3.1. Conceptualizing the problem and the associated uncertainties

We have chosen to focus in this case study on discovering plausible undesirable dynamics of the copper price. What is deemed undesirable differs from one actor to another. Broadly speaking, periods of high prices are undesirable for consumers, periods of low prices are undesirable to producers, and very rapid changes in price are undesirable to both. Later, in analyzing the clusters of dynamics, we will return to these forms of undesirability in order to characterize the dynamics of the identified clusters.

The dynamics of the copper price arise out of an interaction between supply and demand. The supply side of the copper industry, and in particular the supply chain, is extensively documented [17,18]. The chain starts with either the amount of the resource available or the resource base, although there is some discussion about which of these two to use in estimating the maximum amount of available copper [19–21]. The available resources are translated into reserves and a reserve base. The reserves are the fraction of the available resources that can be mined economically; the remainder is the reserve base [36,37]. Part of the reserve base can become part of the reserves due to copper price and technological changes. Copper reserves are transformed into copper via mining, smelting, and refining. Different mining techniques have associated cost structures. The process of smelting and refining depends on the type of copper ore, creating two final products: cathode copper and ingots. Detailed descriptions for the mining, smelting, and refining of copper, and their available capacity are readily available in the literature [17,18]. With respect to the copper ore that is being mined, the ore grade is declining [17], and, as a consequence, the marginal costs of copper mining in terms of energy demand for refining (lower grade) copper are increasing [38]. How this decline of the ore grade will evolve in the future is uncertain.

With respect to the demand side of the copper system, the uncertainties are much more profound. There is an ongoing debate about why the copper price has been so high. Two main explanations that are offered are economic growth in emerging economies and energy transitions [16,17]. These explanations reflect a different idea about how copper demand arises. It can be seen as a function of world population and GDP, or as a function of different usages of copper. Apart from uncertainties related to demand, there are also deep uncertainties related to the future development of energy prices and the dynamics of substituting copper with other materials. The future energy price affects both the copper price as well as the price of substitutes. In particular, the price of aluminum, a major substitute for copper in electronic applications, is highly dependent on energy price developments. In addition, the dynamics of substitution depend on the available substitutes, the amount of materials needed in case of substitution, and the price of the substitutes in comparison to the copper price [24], each of which can change over time.

3.2. The ensemble of models

The models used for generating future scenarios were constructed using system dynamics (SD) modeling [39] and were implemented in Vensim [40]. These models have been designed in such a way that they allow for the exploration of the specified uncertainties, while offering an endogenous explanation for the overall behavior of the copper system in terms of how supply, demand, price, and substitution change over time. To cope with uncertainty about whether demand is a function of population and GDP (i.e. the top down perspective) or whether demand is a function of usages of copper (i.e. the bottom up perspective), we developed two distinct SD models. These two models have a common core with respect to the supply of copper but differ with respect to how demand is modeled. Below we offer a concise description of both models and their common core. For a more detailed description, see [41].

3.2.1. Common core: supply

The copper supply is modeled as a stock-flow structure, from resource base, resources, and reserve base via mining and refining, to the global consumption of refined copper and copper use. When global copper in use has reached the end of its lifetime, it is partially collected via the global secondary copper market as scrap and is recycled as copper recovered from scrap. These stocks and flows constitute the physical and technical backbone of the system [17,42]. With respect to the discussion about resource versus resource base [19,20], we assume that the marginal cost of copper is a function of the amount of energy needed to mine it and the price of energy. The price is a function of demand and supply. The difference between the marginal cost and the price ultimately defines how much copper can be mined. The structure of resources and reserves further largely follows the McKelvey classification [37]. The discovery of new resources is modeled through the semi-autonomous findings of independent exploration or junior companies.

The mining, smelting, and refining capacities are modeled as stocks. Capacity can exist in one of three states: under construction, in use, and mothballed [43]. During poor economic times, capacity is first mothballed and, only after continued losses, decommissioned. Further, due to the maturity of the copper market, learning effects are assumed to be negligible in relation to the increasing costs related to the declining ore grade. No separate capacity for recycling is modeled, since the recycling of copper uses the same smelter and refinery capacity as newly mined copper.

3.2.2. Top down demand

The causal loop diagram (CLD) [39,44] in Fig. 1 shows the main feedback loops of the top down demand sub model at a high aggregation level. In this model, the intrinsic demand is caused by the development of the world population, the development of GDP per capita, and the effect of GDP on the demand for copper. This model allows for exploring the effects of both the growing world population on copper demand and, potentially even more important, the effects of the increase in wealth of the large populations of emerging economies.

3.2.3. Bottom up demand

A high level CLD of the bottom up demand sub model is shown in Fig. 2. In this model, intrinsic demand is a function of the various uses of copper. Following [45], we define six main uses for copper, all of which can have a quasi-independent development over time. Moreover, substitution is based on each of the uses of copper and can behave differently over time for each use. This model allows for exploring, for example, the effects of energy transitions, which among others affect the use of copper in cars [45] and the need for new energy infrastructure to connect renewable energy sources like wind parks and solar plants [16]. This model has a significantly larger number of uncertain parameters than the top down demand model.

3.2.4. Matching supply and demand

Copper supply (available global inventories of refined copper) and demand (intrinsic demand minus substitution) determine the copper price relative to the marginal costs. The copper price influences the total copper demand in two ways: directly, by price elasticity effects on demand, and indirectly by substitution of copper demand (Fig. 3). Price and substitution effects both have a short-term and long-term effect on the price. On average, the long-term effect is larger than the short-term effect. When

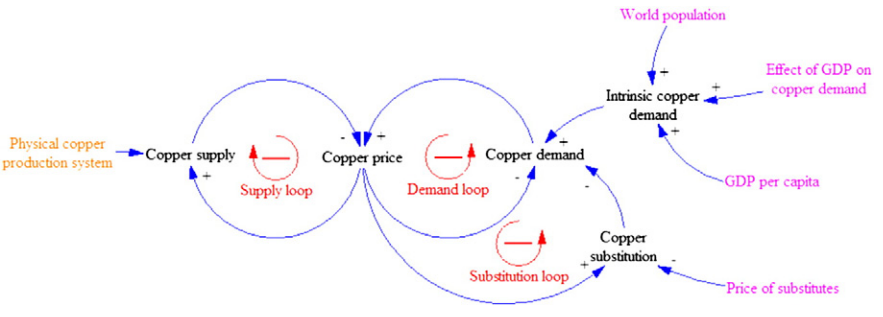


Fig. 1. High level CLD of top down model. Arrows indicate a causal influence and the sign indicates the direction of the influence. The circular arrows with a sign in the middle represent feedback loops. The sign indicates whether the loop is positive or negative.

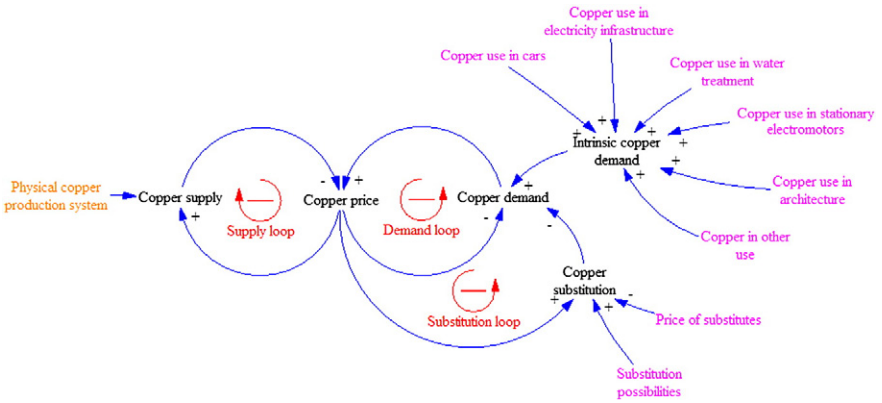


Fig. 2. High level CLD of bottom up model. Arrows indicate a causal influence and the sign indicates the direction of the influence. The circular arrows with a sign in the middle represent feedback loops. The sign indicates whether the loop is positive or negative.

the total demand cannot be fulfilled, the unfulfilled demand is accumulated. This accumulated unfulfilled demand will decline over time, as long as copper scarcity persists. The actual demand for refined copper is thus the demand formed by intrinsic demand, price effects, substitution, and accumulated unfulfilled demand.

3.3. The design of experiments

In order to explore the behavior of the two models over the uncertainties, a shell written in Python was utilized [46]. This 'EMA workbench' can control Vensim through its Dynamic Link Library [47]. The workbench is responsible for generating input values

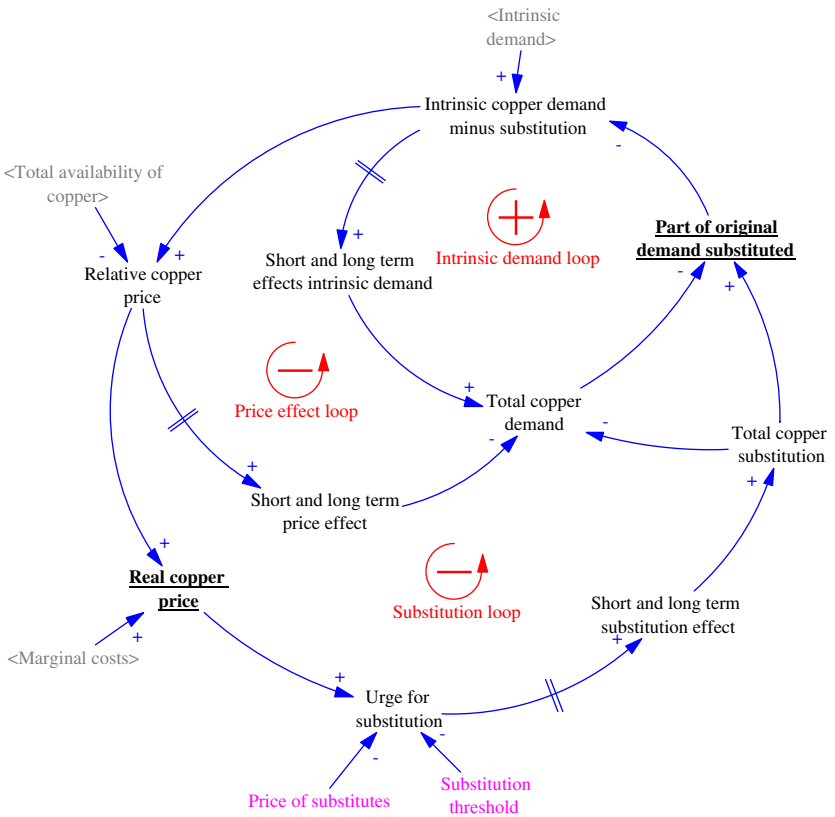


Fig. 3. CLD of the matching of demand to the price. Arrows indicate a causal influence and the sign indicates the direction of the influence. Arrows with a strike through indicate delayed causal influences. The circular arrows with a sign in the middle represent feedback loops. The sign indicates whether the loop is positive or negative.

for the various uncertainties, setting these values in the Vensim models, executing the Vensim models, and storing the results. The workbench supports parallel processing, reducing the required computational time.

The top down model contains 31 uncertain factors, and the bottom up model contains 57 uncertain factors. In order to compare the extent to which the way of modeling demand has an impact on the results, we first identified the uncertainties that are shared by the two models. We generated a Latin Hypercube sample containing 2500 experiments for these 12 shared uncertainties. Next, for each model, the uncertainties unique to that model were identified, and a Latin Hypercube was generated for these uncertainties. By combining these two sets of Latin Hypercube samples for each model, a complete set of experiments for each model is derived. This results in a dataset containing 5000 computational experiments.

3.4. Analysis of results

3.4.1. Time series clustering

In order to identify the types of dynamics present in the ensemble of model results, we use an agglomerative hierarchical clustering approach with a specifically tailored metric for identifying similar dynamics. The dynamic behavior over time can be understood as being a concatenation of atomic behavior patterns [48]. The atomic behavior mode is based on the sign (positive, negative, and zero) of the slope and curvature. Thus nine atomic behavior modes exist. Each time series is transformed into a concatenation of atomic behavior modes. Next, similarity between two time series is based on comparing these concatenations. The distance between two dynamics is then the average deviation across the entire concatenation. This approach is essentially an extension of the behavior pattern features discussed in [49]. See [50] for a more detailed elaboration on this similarity metric.

Fig. 4 shows the dendrograms [51] resulting from the agglomerative clustering with the behavioral distance metric. We have shown only the top four levels. For explorative purposes, we chose to extract 30 clusters and review their dynamics. This offers a balance between choosing a few clusters and potentially missing insight into the variety of possible dynamics generated by the 5000 experiments, and choosing too many clusters resulting in information overload. In order to review the dynamics of each cluster, we identified the computational experiment that had the lowest average distance to the other experiments in the same cluster and used it as the exemplar for the dynamics in that cluster. In this way, we are able to move from 5000 computational experiments to 30 exemplar dynamics.

Fig. 5 shows these 30 exemplars. To ease the interpretation of the plots, we plotted only 10 exemplars jointly, and scaled the values so that they all have the same starting point. The y-axis is shared across the plots. In the top and bottom plot we see exemplars that show a slow growth of the copper price over time. We also observe several exemplars that show sharp price changes.

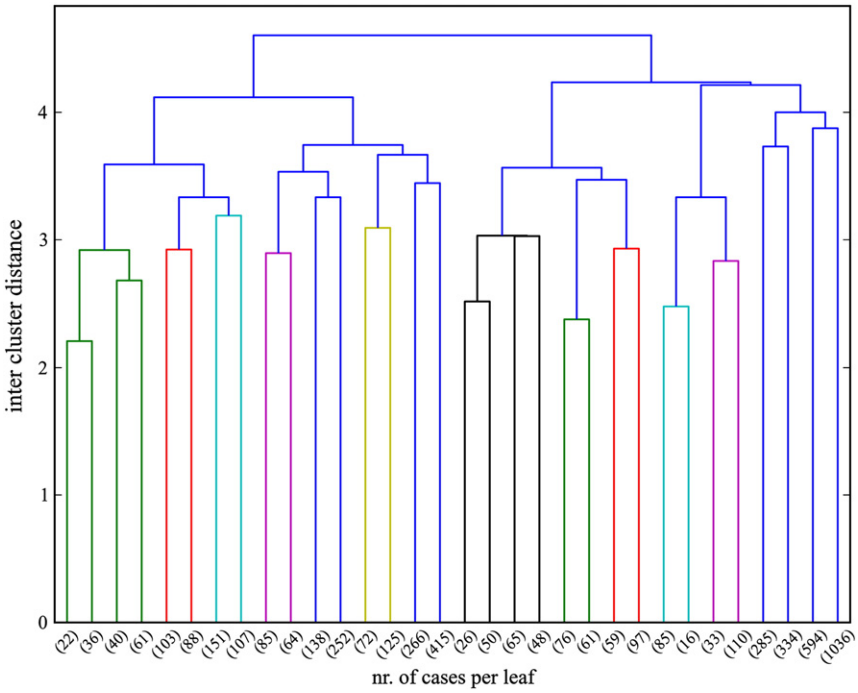


Fig. 4. Dendrogram of the clustering of the dynamics of the copper price. The y-axis specifies the intercluster distance, and the x-axis indicates the number of computational experiments allocated to the lowest level in the dendrograms.

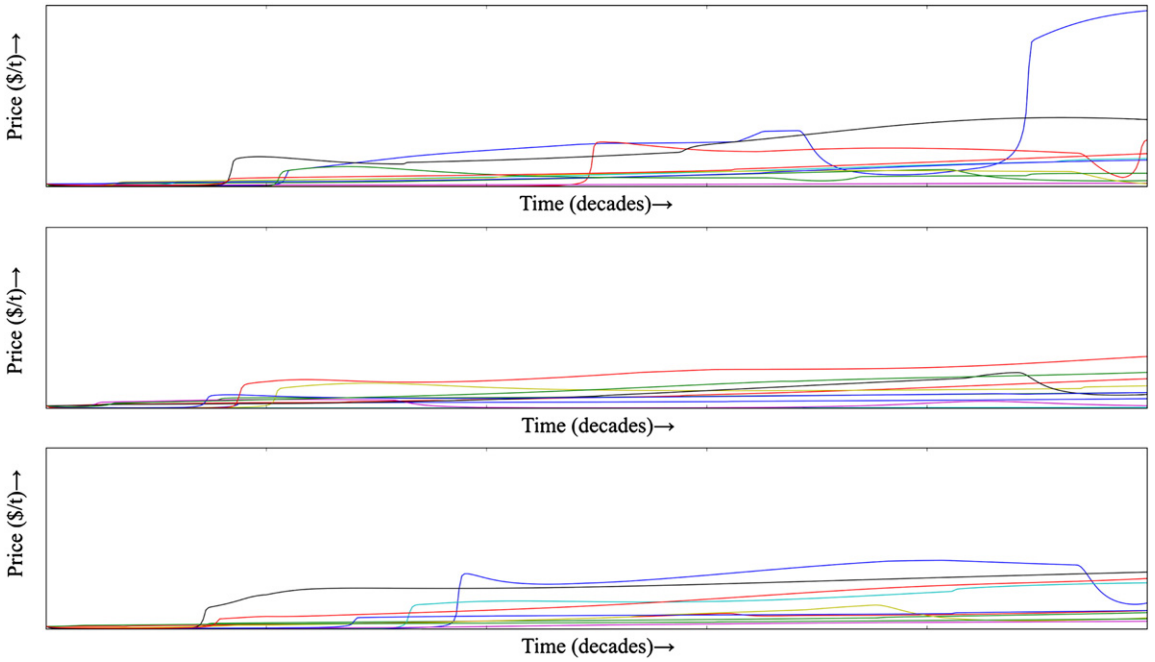


Fig. 5. Thirty exemplars of the dynamics of the copper price.

3.4.2. Scenario discovery

Having identified the behavioral landscape through time series clustering, the next step is to specify which of the identified dynamics are undesirable. We assume that high prices are undesirable for consumers, low prices are undesirable to producers, and (rapid) price changes are undesirable to both. The time series clustering reveals several clusters with sharp price fluctuations. For illustrative purposes, we see regions in the behavioral landscape that show sharp changes in price as the regions of interest. Other criteria could also be used [see e.g. 52,53]. To identify these regions, we use a simple rule. If the absolute value of the first order derivative of the price is higher than the 2 anywhere over the run, the run is classified as a run of interest. Next, we use PCA-PRIM [34] to identify where in the uncertainty space these undesirable dynamics originate. This choice for PCA-PRIM was due to the fact that without the PCA preprocessing step, PRIM failed to identify subspaces in the uncertainty space from which the regions of interest in the behavioral landscape originate. Given that the experiments for both models are different, we applied PCA-PRIM to both sets of experiments separately.

3.4.2.1. Top down model. For the top down model, PRIM identified two boxes which jointly contain 55% of the experiments of interest. These two boxes are primarily characterized by six restricted principal components. Table 1 shows the loadings of these six principal components on the uncertain parameters. These loadings indicate the strength and direction of the relation between the principal component and the uncertainty. For example, 'minimum usage smelting and refining capacity' is strongly positively related to the 'Supply_3' principal component, while 'initial value GDP per capita' is negatively related to the 'Demand_6' principal component. As can be seen, we clustered the uncertain parameters into parameters related to the capacity subsystem, the demand and the supply subsystems, and the price subsystem.

With respect to capacity, the results indicate that the main driver for undesirable price fluctuations coming from the capacity subsystem are the permit time for additional smelting and refining capacity, the initial value for the smelting and refining capacity, and the lifetime of the mines. The underlying dynamic here is that, if there is a misbalance between the mining capacity and the refining and smelting capacity, there can be a buildup of raw copper. But, due to a lack of smelting and refining capacity and the fact that it can take a long time to expand this capacity, it takes a very long time before there is sufficient refining capacity. The long permit time results in an overshoot of new smelting and refining capacity, thus resulting in a sudden increase of refined copper becoming available on the market. This in turn, triggers a rapid drop of the price. Because of the overshoot in capacity, the stock of raw copper is depleted, reducing utilization of the capacity, in turn causing the mothballing of this capacity, and potentially even its dismantling. This effect of the capacity system is aggravated by the supply subsystem, which contributes to price fluctuations if all or a large fraction of the mining and smelting capacity is continuously used.

The demand subsystem contributes to undesirable price fluctuations if the economy is shrinking and the initial value for GDP per capita is low. Intrinsic demand for copper is highly sensitive to small changes in GDP (this is essentially the meaning of 'normalization value GDP') [54]. That is, copper demand rapidly decreases. However, historically the expectation is that demand will continue to increase (represented by the 'initial value long term effect intrinsic demand'), and it takes a while before this expectation is changed ('long-term increased demand due to intrinsic demand'). Hence, in case of a falling intrinsic demand and the expectation of continuing growth of demand, the demand subsystem contributes to price fluctuations.

Table 1
Rotation matrix for the top down model. The columns are the principal components. The values indicate the loadings of the uncertainties on these components. Loadings indicate the weight by which the normalized value for an uncertainty needs to be multiplied in order to get the principal component score. Thus, the higher the loading, the stronger the relation, and the sign indicates the direction of the relation. Only the principal components that are most important according to the PRIM results are shown. Irrelevant principal components and their related uncertainties have been removed.

	Capacity_4	Demand_5	Demand_6	Demand_7	Supply_3	Price_1
Average smelting and refining capacity permit term	0.59					
Initial mining capacity	−0.22					
Initial mining capacity in preparation	−0.19					
Initial smelting and refining capacity	0.50					
Average mine lifetime	0.55					
Initial value GDP per capita		−0.16	−0.81	−0.41		
Initial value long term effect intrinsic demand		0.43	−0.20	0.05		
Long term increase demand due to intrinsic demand		−0.44	0.08	0.07		
Base economic growth		−0.72	0.13	0.05		
Amplification factor of intrinsic demand		−0.05	−0.23	0.11		
Amplification factor of relative price effect		−0.10	0.13	0.03		
Normalization value GDP		−0.02	−0.36	0.90		
Long term copper price elasticity		0.25	0.29	0.03		
Minimum usage smelting and refining capacity					0.58	
Initial inventories of refined copper					−0.06	
Switch between using all capacity, or capacity based on forecast					−0.80	
Forecasting time horizon					−0.11	
Price amplification factor						−0.86
Production time						0.51

Table 2
Rotation matrix for the bottom up model. The columns are the principal components. The values indicate the loadings of the uncertainties on these components. Loadings indicate the weight by which the normalized value for an uncertainty needs to be multiplied in order to get the principal component score. Thus, the higher the loading, the stronger the relation, and the sign indicates the direction of the relation. Only the principal components that are most important according to the PRIM results are shown. Irrelevant principal components and their related uncertainties have been removed.

	Supply_3	Demand_15	Recycling_10	Price_1
Forecasting time horizon	−0.56			
Minimum usage smelting and refining capacity	0.72			
Part of resource base (sea based)	0.39			
Threshold for junior companies to start deep sea reserve base development	0.13			
Goal for copper in infrastructure 2050		0.13		
Initial value GDP per capita		−0.39		
Number of cars in 2000		0.09		
Cu in vehicles BEV		0.14		
Long term effect on demand period		0.37		
Long term increase demand due to intrinsic demand		−0.28		
Base economic growth		0.17		
Initial value long term effect price		0.45		
Amplification factor of intrinsic demand		-0.19		
Switch new cars		0.21		
Cu in vehicles cityBEV		0.02		
Long term effect intrinsic demand period		−0.21		
Switch World population		−0.12		
Amplification factor of relative price effect		−0.04		
Long term copper price elasticity		−0.43		
Growth of copper in architecture		−0.17		
Collection rate copper products (water treatment)			0.02	
Average lifetime of copper in use (architecture)			−0.02	
Collection rate copper products (automotive)			−0.04	
Percentage copper recovered from scrap			−0.24	
Initial value global copper in scrap (stationary electromotors)			−0.06	
Initial use grade for water treatment			−0.02	
Initial total copper in use			0.95	
Average lifetime of copper in use (other uses)			−0.14	
Copper score during treatment (water treatment)			−0.03	
Copper score during treatment (energy infrastructure)			−0.05	
Initial value global copper in scrap (other uses)			−0.07	
Price amplification factor				−0.92
Production time				0.39

The price subsystem contributes to price fluctuations if the relative price of copper is highly sensitive and the production time is low. The production time affects the time it takes for the inventories of copper to progress through the market. The lower this value, the more the amount of copper fluctuates, which in turn is aggravated by the 'price amplification factor'.

3.4.2.2. Bottom up model. For the bottom up model, PRIM identified two boxes that jointly contain almost 50% of the experiments of interest. These two boxes are primarily characterized by four principal components, which are related to supply, demand, recycling, and price (Table 2). The contribution of the price subsystem to the price fluctuations is identical to its contribution in the top down model. In contrast to the top down model, there are no principal components related to the capacity subsystem. The supply subsystem contributes to price volatility if the time horizon used in forecasting future demand levels is quite short in combination with a high value for the minimum usage of smelting and refining capacity. The demand subsystem contributes to price volatility if the price elasticity is low, the long term effects of demand and price are quite strong, and the initial value for GDP per capita is low. This corresponds to an inert demand sector that only slowly adjusts to price changes. The recycling subsystem contributes to price volatility if the initial amount of copper in use is quite high. This means that, over the simulation, a large amount of copper suddenly becomes available via recycling. This lowers the price, but demand responds only slowly, resulting in a prolonged period of low prices. During this period, mining capacity is mothballed and potentially removed. By the time demand starts to respond to the low price, mining capacity and smelting and refining capacity have been reduced, thus resulting in a situation with more demand than supply, driving up the price.

4. Discussion

The aim of this paper was to illustrate how scenario discovery can be used to develop dynamic scenarios. To this end, we utilized a behavioral time series clustering algorithm to identify the behaviors of interest. Next, PCA-PRIM was used to identify the causes for the behaviors of interest. Looking at the final results, it appears that such a technique is able to identify a subset of uncertainties that produce a particular dynamic. In the case of the top down model, we started with 31 uncertain factors, and we were able to identify a substantially smaller subset of uncertainties that is responsible for a particular type of undesirable behavior. Similarly, in case of the bottom up model, we started with 57 uncertainties and were able to reduce it also to a substantially lower number of uncertainties responsible for generating the undesirable dynamics. Still, in both cases, only around 50% of the cases of interest have been accounted for. It thus appears that the dynamics of the model are due to a very particular dependency among the uncertainties. So, the idea of tracing back the behavior of interest to a region of the model input space does not hold for all of the regions of interest.

The identified scenarios suggest that given the current functioning of the copper system, sharp price fluctuations are indeed a plausible future. That is, it is conceivable that the prices will come down quickly and stay there for a period of time, before rising sharply again. Thus, the assumption that the current high price level is necessarily here to stay is not justified in light of these results. With respect to the identified undesirable dynamics and their underlying causes, there are several policy implications. There is a mismatch between the production of raw copper from mining and the production of copper in scrap through recycling and the smelting and refining capacity in the top down model, and to a lesser extent in the bottom up model. This mismatch results in oscillations in the supply of refined copper. To counteract these oscillations, there is a need for supply chain management. Another policy implication is that given the uncertainty related to future demand, and the impact of a declining demand on the price in case of the bottom up model, there is a need to create flexibility in the development of new mining capacity and new smelting and refining capacity. The use of real options or outside the box thinking with respect to the identification of new reserves is the way of addressing this need for flexibility [43,55]. A last important policy implication is that price oscillations can be counteracted through the establishment of strategic reserves that are used in a counter cyclical way. Thus, copper is bought when prices fall below a particular level and sold when the prices rise above a particular level.

To improve the interpretability of the results, we experimented with various techniques. First, we decided to use the time series clustering only as a means of interpreting the model outcomes and derive from it a rule for identifying the cases of interest, rather than using the clusters themselves as cases of interest. This improved the results substantially. Second, we used a preprocessing step based on principal components analysis prior to applying PRIM. Where PRIM without this preprocessing step was unable to identify a region in the model input space, PCA-PRIM was able to identify regions. These regions identified by PCA-PRIM are in the eigenspace rather than in the model input space, thus necessitating an additional interpretation step based on the loadings of the model input parameters on the principal components — that is, the dimensions that span the eigenspace.

One way for further improving the coverage, the fraction of all the cases of interest in a single prim box, would be to modify the objective function used in PRIM. In the current version, the objective function aims at maximizing the mean of the cases inside a box divided by the change of the cardinality of the cases inside the box. Assuming that a binary classification is used to identify cases of interest, this is the same as maximizing the density – the fraction of cases of interest in the box divided by the total number of cases in the box [10] – of the box. However, it is possible to modify the algorithm to also take into account the change in coverage — the fraction of cases of interest in the box divided by the total number of cases of interest [10]. That is, one could select the box that increases the density the most relative to the decrease in coverage. This way, the algorithm would search for boxes that have both a relatively high coverage and relatively high density. Conceptually, this is closely related to a modification of PRIM known as 'flexible PRIM', where one can make a tradeoff between mass and mean [33]. A related way in which the results could be improved is by adding some form of penalty to the objective function based on the number of restricted dimensions. Ideally, one would like to identify a small manageable subset of the uncertain factors that is responsible for a large number of the

undesirable dynamics. In the approach outlined by [7,13], this is handled by making the PRIM process interactive. A user can then make his or her own tradeoff between increasing density and the number of restricted dimensions. This increases the interpretability of the results, but it makes reproducibility more difficult. Moreover, it introduces some form of expert judgment into the procedure. However, it would be convenient if PRIM were able to be used in an automatic mode, where the tradeoffs between density, coverage, and the number of restricted dimensions are pre-specified and reproducible.

Another direction for improving the results is in the time series clustering. We used a behavioral metric introduced in [50]. This metric helped in identifying plausible dynamics. However, as indicated in [50], the metric is less suited to oscillatory dynamics. Moreover, a more fine grained analysis of the types of dynamics could further improve the results. So far, we have lumped all cases of sharp changes in price together. A more fine grained approach could be to separate sharp price fluctuations that become worse over time from those cases where the price fluctuations die out.

A final direction for improving the results could be through model simplification. The two models used in this case are quite large and describe the copper system in detail. A simplified model at a higher level of aggregation, which still contains the most elemental feedback loops and the supply chain stock flow structures, could conceivably produce results very similar to the results reported here. However, by producing these results with a simpler model, the number of uncertainties over which the exploration is done, and the interpretation of their results, would be reduced substantially. Moreover, model simplification could help in explaining and communicating better the results to decision-makers and other stakeholders in the copper system.

5. Conclusions

Scenarios are a powerful and frequently employed means for communicating and characterizing uncertainty in decision making. The literature on scenarios and approaches for the development of scenarios is wide and diverse. Various challenges and issues are acknowledged in this literature, including how to cope with actors with quite diverse worldviews, how to develop scenarios when there is a lack of consensus or agreement between stakeholders, and how to best represent the richness and multiplicity of plausible futures in a finite and small set of scenarios. Scenario discovery is a technique that can be employed to cope with some of these challenges. In particular, scenario discovery can be used to identify a small set of scenarios of interest given a very large and diverse set of uncertainties. Scenario discovery also has the potential to be of use if there is a lack of consensus or a disagreement among the actors, thus helping in the development of scenarios in the presence of deep uncertainty. Scenario discovery could also be used to identify a small set of scenarios that is fully representative, both in terms of behavior and in terms of origins in the multi-dimensional uncertainty space, for a large ensemble of scenarios or specific subsets of the large ensemble.

In the literature on scenario discovery, scenarios have been treated as future states of the world in which a candidate policy was performing poorly [7,10,11]. The future states of the world were generated by a single model with a few uncertain parameters over which a parameter sweep was executed. We extended this approach in multiple ways. First, we broadened the notion of scenarios in the context of scenario discovery to cover any region of the model outcome space that is of interest for some reason. Moreover, we were interested in the dynamics over time instead of a future state of the world. The introduction of dynamics over time necessitated the need for the clustering of time series based on their dynamic behavior pattern, thus allowing the use of the same machine learning techniques that had already been used successfully in the context of scenario discovery. Second, we applied scenario discovery to the case of copper scarcity, where there was deep uncertainty about how to model an aspect of the system, resulting in the development of two distinct models. Moreover, these two models jointly cover a significantly larger number of uncertainties than previously reported applications of scenario discovery [7,10].

In the copper case, we used a time series clustering approach to identify behaviorally distinct clusters. In light of the identified behaviors, we specified a rule for classifying results as being of interest or not. For both models, we identified where in the model input space these behaviors of interest originated. We showed that undesirable price fluctuations can result from a mismatch of capacity in the copper production chain. In case of the bottom up model, this creates problems if there is a shock to the system due to a rapid increase of copper scrap due to recycling. Another important contributing factor to undesirable fluctuations of the price is the situation where copper demand starts to drop, while the expectation remains that copper demand will grow. The case showed how scenario discovery can be used to identify a small subset of uncertain factors that is responsible for a particular dynamic over the time period of interest. However, there is room for improvement of the results. In particular, the approach for identifying the combinations of uncertainties responsible for a particular dynamic can be improved, for example by fine tuning it in such a way that it searches for a small set of uncertainties that have both a high coverage and high density.

References

[1] L. Börjeson, M. Höjer, D. K.H., T. Ekvall, G. Finnveden, Scenario types and techniques: towards a user's guide, Futures 38 (2006) 723–739.
[2] R. Bradfield, G. Wright, G. Burt, G. Cairns, K. van der Heijden, The origins and evolution of scenario techniques in long range business planning, Futures 37 (2005) 795–812.
[3] European Environmental Agency, Looking Back on Looking Forward: A Review of Evaluative Scenario Literature, European Environmental Agency, Luxembourg, 2009.
[4] S.A. van 't Klooster, M.B.A. van Asselt, Practising the scenario-axes technique, Futures 38 (2006) 15–30.
[5] P. Goodwin, G. Wright, The limits of forecasting methods in anticipating rare events, Technol. Forecast. Soc. Chang. 77 (2010) 355–368.
[6] S. Popper, J. Griffin, C. Berrebi, T. Light, E.Y. Min, Natural Gas and Israel's Energy Future: A Strategic Analysis Under Conditions of Deep Uncertainty, in: RAND, Santa Monica, California, 2009.

[7] B.P. Bryant, R.J. Lempert, Thinking inside the box: a participatory computer-assisted approach to scenario discovery, Technol. Forecast. Soc. Chang. 77 (2010) 34–49.
[8] P. Schwartz, The Art of the Long View, John Wiley & Sons, Chichester, England, 1991.
[9] K. van der Heijden, Scenarios: The Art of Strategic Conversation, Wiley, 1996.
[10] D.G. Groves, R.J. Lempert, A new analytic method for finding policy-relevant scenarios, Glob. Environ. Chang. 17 (2007) 73–85.
[11] R.J. Lempert, D.G. Groves, S. Popper, S. Bankes, A general analytic method for generating robust strategies and narrative scenarios, Manag. Sci. 52 (2006) 514–528.
[12] S. Bankes, Exploratory modeling for policy analysis, Oper. Res. 4 (1993) 435–449.
[13] R.J. Lempert, B.P. Bryant, S. Bankes, Comparing Algorithms for Scenario Discovery, in: Rand, Santa Monica, 2008.
[14] R.J. Lempert, S. Popper, S. Bankes, Shaping the next one hundred years: new methods for quantitative, in: Long-Term Policy Analysis, RAND, Santa Monica, California, 2003.
[15] Index Mundi, copper, grade A cathode monthly price — US dollars per metric ton, in: Commodities, Metals, 2011.
[16] R. Kleijn, E. van der Voet, Resource constraints in a hydrogen economy based on renewable energy sources: an exploration, Renewable Sustainable Energy Rev. 14 (2010) 2784–2795.
[17] ICSG, The World Copper Factbook. (Lisbon, Portugal) , 2010.
[18] A. Lossin, Copper, in: Ullman's Encyclopedia of Industrial Chemistry, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, 2005.
[19] R.B. Gordon, M. Bertram, T.E. Graedel, On the sustainability of metal supplies: a response to Tilton and Lagos, Resour. Policy 32 (2007) 24–28.
[20] J.E. Tilton, G. Lagos, Assessing the long-run availability of copper, Resour. Policy 1-2 (2007).
[21] R.B. Gordon, M. Bertram, T.E. Graedel, From the cover: metal stocks and sustainability, Proc. Natl. Acad. Sci. 103 (2006) 1209–1214.
[22] D.H. Meadows, J. Richardson, G. Bruckman, Groping in the Dark, John Wiley & Sons, 1982.
[23] B.J. Skinner, A second iron age ahead, Am. Sci. 64 (1976) 259–269.
[24] R.B. Gordon, K. T.C., W.D. Nordhaus, B.J. Skinner, Towards a New Irong Age? Quantitative Modeling of Resource Exhaustion, Harvard University Press, Cambrdige, Massachusetts, 1987.
[25] M. Rademaker, J. Kooroshy, The global challenge of mineral scarcity, in: Enriching the Planet, Empowering Europe — Optimising the Use of Natural Resources for a More Sustainable Economy, The Hague, the Netherlands, 2004.
[26] J.H. Friedman, N.I. Fisher, Bump hunting in high-dimensional data, Stat. Comput. 9 (1999) 123–143.
[27] Y. Ben-Haim, Information-Gap Decision Theory: Decisions under Severe Uncertainty, 2 ed. Wiley, New York, 2006.
[28] J.H. Kwakkel, W.E. Walker, V.A.W.J. Marchau, Classifying and communicating uncertainties in model-based policy analysis, Int. J. Technol. Policy Manage. 10 (2010) 299–315.
[29] T.W. Liao, Clustering of time series data — a survey, Pattern Recognit. 38 (2005) 1857–1874.
[30] S. Theodoridis, Pattern recongition, in: Encyclopedia of Information Systems, 2003, pp. 459–479.
[31] J. Han, M. Kamber, Data Mining: Concepts and Techniques, Morgan Kaufmann, San Francisco, 2001.
[32] E. Keogh, S. Kasetty, On the need for time series data mining benchmarks: a survey and empirical demonstration, Data Min. Knowledge Discov. 7 (2003) 349–371.
[33] I.G. Chong, C.H. Jun, Flexible patient rule induction method for optimizing process variables in discrete types, Expert Syst. Appl. 34 (2008) 3014–3020.
[34] S. Dalal, B. Han, R. Lempert, A. Jaycocks, A. Hackbarth, Improving Scenario Discovery using Orhogonal Rotations, Environmental Modelling & Software, (under review).
[35] E.W. Weisstein, Covariance matrix, in: MathWorld — A Wolfram Web Resource, 2012.
[36] JORC, The JORC Code — Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves, JORC, 2004.
[37] V.E. McKelvey, Mineral resources estimates and public policy, in: United States Mineral Resources, Geological Survey Professional Paper, 820, 1973, pp. 19–20.
[38] B.X. Sun, Z.R. Nie, Y. Liu, Z.H. Wang, X.Z. Gong, Exergy-based model of the depletion of mineral resources, Mater. Sci. Forum 650 (2010) 1–8.
[39] J.D. Sterman, Business Dynamics: Systems Thinking and Modelling for a Complex World, McGraw-Hill, Boston, 2000.
[40] Ventana Systems Inc., Vensim Reference Manual, in: , 2011.
[41] W.L. Auping, The uncertain future of copper: An Exploratory System Dynamics Model and Analysis of the Global Copper System in the Next 40 Years, Delft University of Technology, Delft, 2011.
[42] Wiley-VCH, Copper, in: Wiley-VCH (Ed.), Ullmann's Encyclopedia of Industrial Chemistry, Wiley, 2003.
[43] S.A. Abdel Sabour, R. Poulin, Mine expansion decision under uncertainty, Int. J. Min. Reclam. Environ. 24 (2010) 340–349.
[44] D.C. Lane, The emergence and use of diagramming in system dynamics: a critical account, Syst. Res. Behav. Sci. 25 (2008) 3–23.
[45] G. Angerer, A. Mohring, F. Marscheider-Weidemann, M. Wietschel, in: Kupfer für Zukunftstechnologien, Fraunhofer ISI, Karlsruhe, 2010, p. 46.
[46] G. van Rossum, Python Reference Manual, CWI, 1995.
[47] Ventana Systems Inc., Vensim DSS Reference Supplement, in: Ventana Systems, Inc., 2010.
[48] D.N. Ford, A behavioral approach to feedback loop dominance analysis, Syst. Dyn. Rev. 15 (1999) 3–36.
[49] G. Yücel, Y. Barlas, Automated parameter specification in dynamic feedback models based on behavior pattern features, Syst. Dyn. Rev. 27 (2011) 195–215.
[50] G. Yücel, A novel way to measure (dis)similarity between model behaviors based on dynamic pattern features, in: The 30th International Conference of the System Dynamics Society, St. Gallen, Switzerland, 2012.
[51] R. Sibson, SLINK: an optimally efficient algorithm for the single-link cluster method, Comput. J. 16 (1973) 30–34.
[52] J.W.G.M. van der Pas, V.A.W.J. Marchau, W.E. Walker, G.P. Van Wee, B. Agusdinata, Exploratory MCDA for handling deep uncertainties: the case of intelligent speed adaptation implementation, J. Multicriteria Decis. Anal. 17 (2010) 1–23.
[53] D. McInerney, R. Lempert, K. Keller, What are robust strategies in the face of uncertain climate threshold responses, Climate Change 112 (2012) 547–568.
[54] H. Wouters, D. Bol, Material scarcity, Stichting Materials Innovation Institute (M2i), 2009.
[55] A.D. Williams, D. Tapscott, Wikinomics: How Mass Collaboration Changes Everything, 2006.

Jan Kwakkel received a Ph.D. from Delft University of Technology. His researched focused on the treatment of uncertainty in long-term airport planning. He currently works as a postdoc on the treatment of uncertainties in model-based decision support for fresh water supply in the Netherlands at Delft University of Technology. Next to this research, he also has an interest in scientometrics and technometrics.

Willem Auping received a M.Sc. from Delft University of Technology. He works as a strategic analyst for the The Hague Centre for Strategic Studies. His research interests are in the area of decision making under deep uncertainty, exploratory modeling and analysis, mineral and metal scarcity, and aging.

Dr. Erik Pruyt is an assistant professor of System Dynamics at Delft University of Technology since 2006, where he and his colleagues develop methods and tools for dealing with deeply uncertain dynamically complex issue. Since 2007, he is also a methodological advisor to the Dutch government regarding National Safety and Security. As such, he co-developed the Dutch National Risk Assessment and Capability Analysis and advised several nations regarding the development of their National Risk Assessments. He also developed methods and software tools for regional risk analysis, and for integrated risk-capability analysis for event-related risks.